



TL;DR

We teach a language model a document by **self-distillation**: the model reads the document and acts as a teacher for itself without it. **SSD** lets the **student write each training rollout** and lets the teacher **take over only at tokens where the two disagree**. It reaches **on-policy** accuracy with **45% fewer supervised tokens** on average and **forgets less** of the model’s general ability.

Background and Motivation

Knowledge in the weights

- Models are often needed on knowledge **absent from pretraining**: private documents, facts that changed after the cutoff, specialised domains.
- Putting it in the prompt costs context and compute **on every query**, and in-context evidence is not always used reliably. Training it into the weights is paid once.

Self-distillation

- Teacher** $\pi_T(\cdot | x, C)$: the model with the document C in context. **Student** $\pi_\theta(\cdot | x)$: the same model without it.
- The student matches the teacher token by token along rollouts y :

$$\mathcal{L}(\theta) = \mathbb{E}_{y \sim \pi_{\text{gen}}} \left[\frac{1}{|y|} \sum_t D(\pi_T(\cdot | x, C, y_{<t}) \| \pi_\theta(\cdot | x, y_{<t})) \right]$$

- Methods differ only in **who writes the rollout**, π_{gen} :
 - off-policy** (teacher writes): fast, but trains on text the student would not produce.
 - on-policy** (student writes, e.g. SDFT): matches test-time behaviour, but slow.

Why On-Policy Training Is Slow

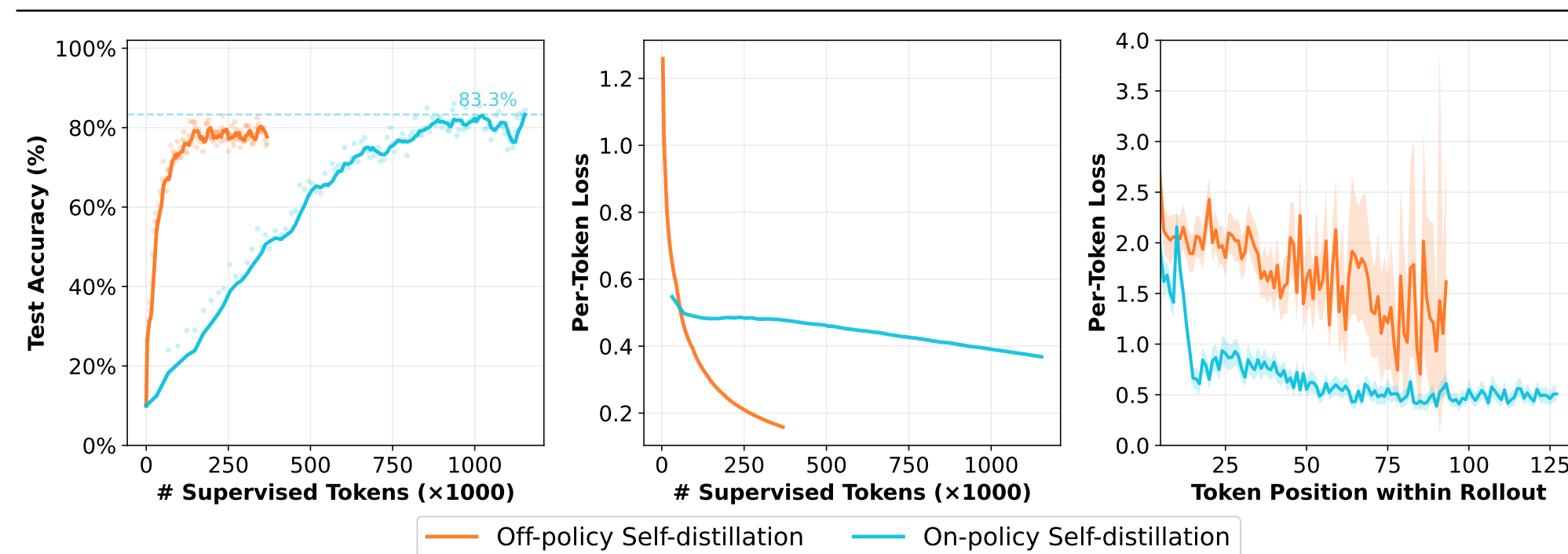


Figure 1. Wikipedia task. Left: accuracy. Middle: per-token loss during training. Right: per-token loss by position within a rollout.

- The per-token loss measures how much the student can learn from a token; we call it the **signal**.
- off-policy** converges $\sim 5\times$ **faster** but plateaus lower.
- Along **on-policy** rollouts the signal **collapses after a few tokens**: the student drifts away from the document, and the teacher, conditioned on that prefix, has little left to correct.

Speculative Self-Distillation (SSD)

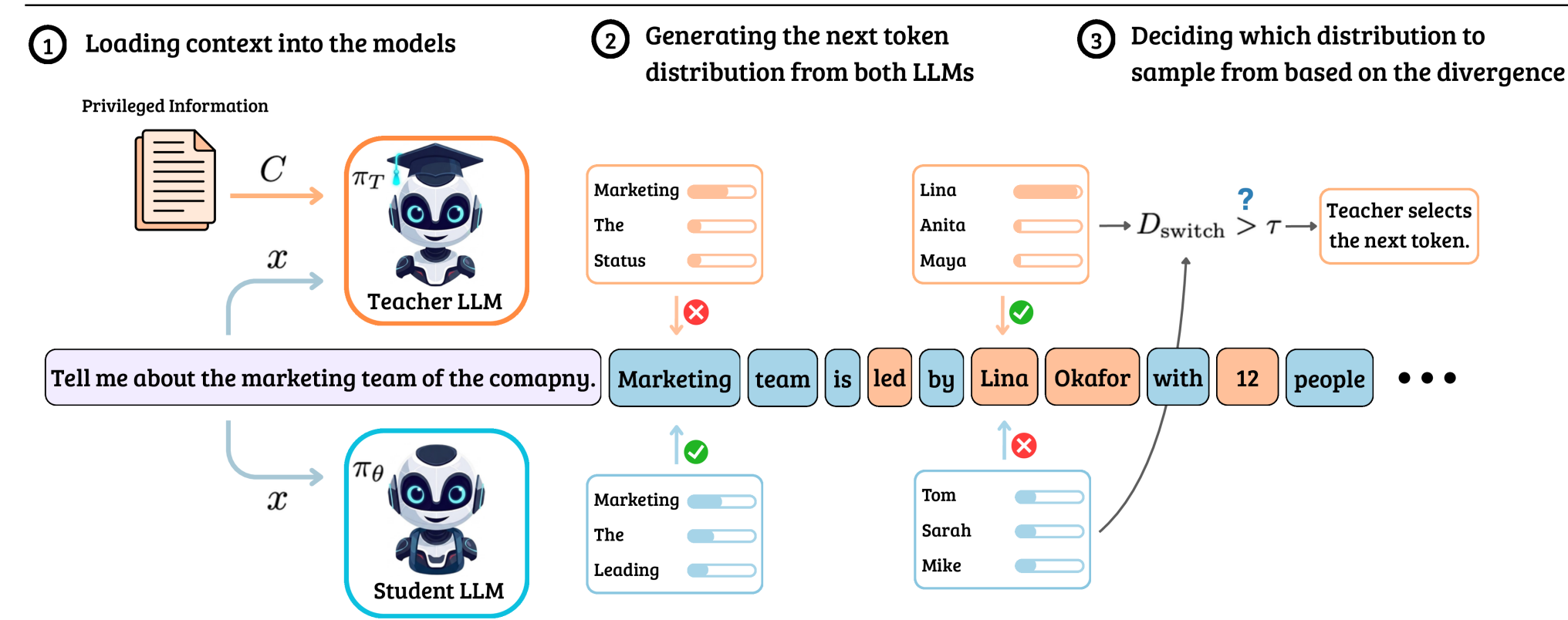


Figure 2. The SSD decoding loop. Both models propose a next-token distribution; when their divergence exceeds τ the teacher picks the token, otherwise the student does.

- The **student generates by default**. At every position we compute $\delta_t = D_{\text{switch}}(\pi_\theta \| \pi_T)$.
- If $\delta_t > \tau$ the position is an **inflection token** and the **teacher samples** it.
- Every position is supervised with the same loss D_{loss} ; only the rollout changes.
- One knob: $\tau \rightarrow 0$ recovers **off-policy**, $\tau \rightarrow \infty$ recovers **on-policy**. With JSD as D_{switch} the divergence is at most $\ln 2$.

Inflection Tokens

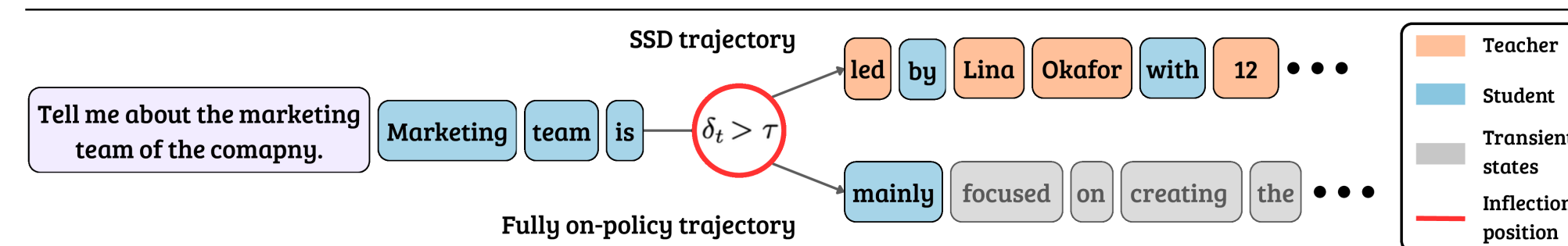


Figure 3. An example rollout. SSD lets the teacher write at the inflection position; on-policy keeps the student’s token and continues into transient states.

- Student and teacher agree on “Marketing team is”, then diverge: the teacher has read the memo (“led by Lina Okafor”), the student has not (“mainly focused on ...”).
- After an **on-policy** inflection, later tokens are **transient states**: the update moves the student away from “mainly”, so it will rarely visit them again and supervision on them is largely wasted.
- SSD** keeps the rollout on the informative path while staying close to the student elsewhere.

Experimental Setup

- Model**: Qwen3-4B-Instruct as both student and teacher; teacher frozen at the initial weights.
- Tasks** (closed-book QA): **Company Memo** (synthetic private document), **Wikipedia** (article after the training cutoff), **Science Q&A** (SciKnowEval Chemistry L-3).
- Baselines**: **off-policy** SFT and FKL on teacher rollouts; **on-policy** SDFT.
- Cost**: supervised tokens, the number of positions the loss is computed on (roughly proportional to compute).

Results

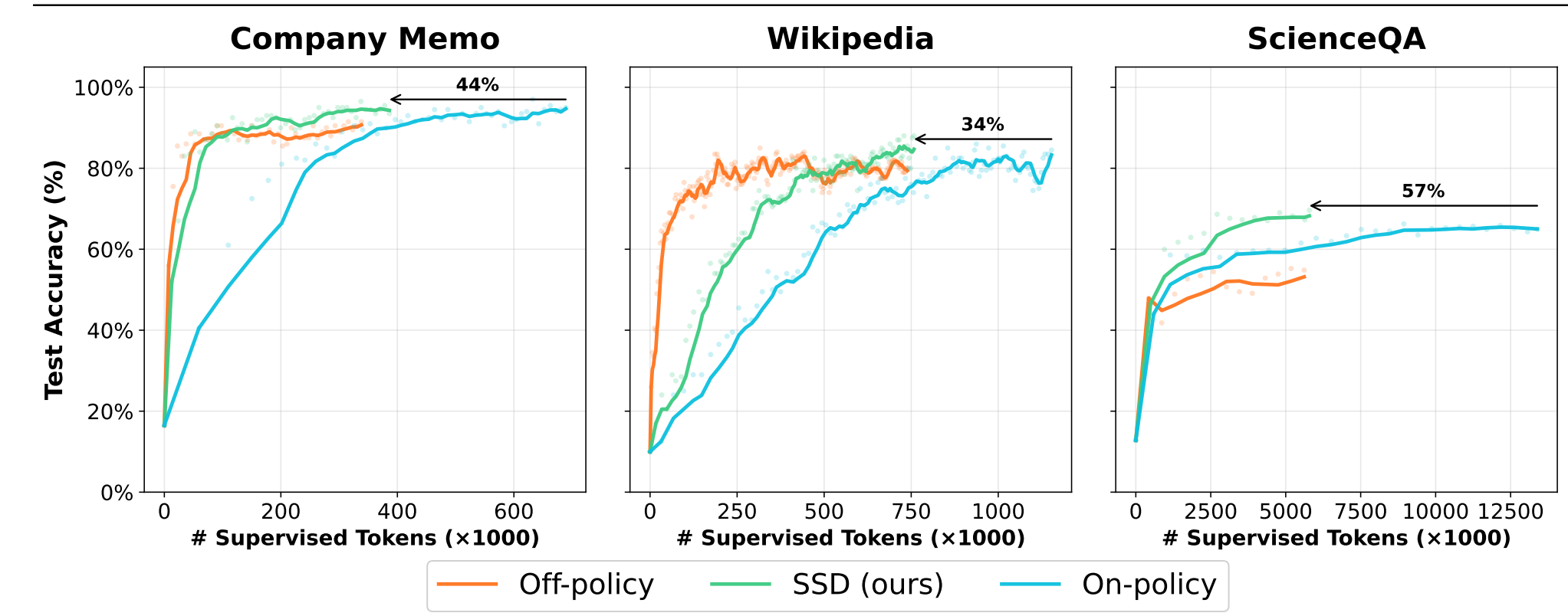


Figure 4. Test accuracy against supervised tokens on the three tasks.

- SSD** rises with **off-policy**’s early slope and continues to the **on-policy** plateau, using **44%, 34% and 57% fewer supervised tokens** than **on-policy**.
- On Science Q&A it ends about **3 points above on-policy** and **15 above off-policy**.

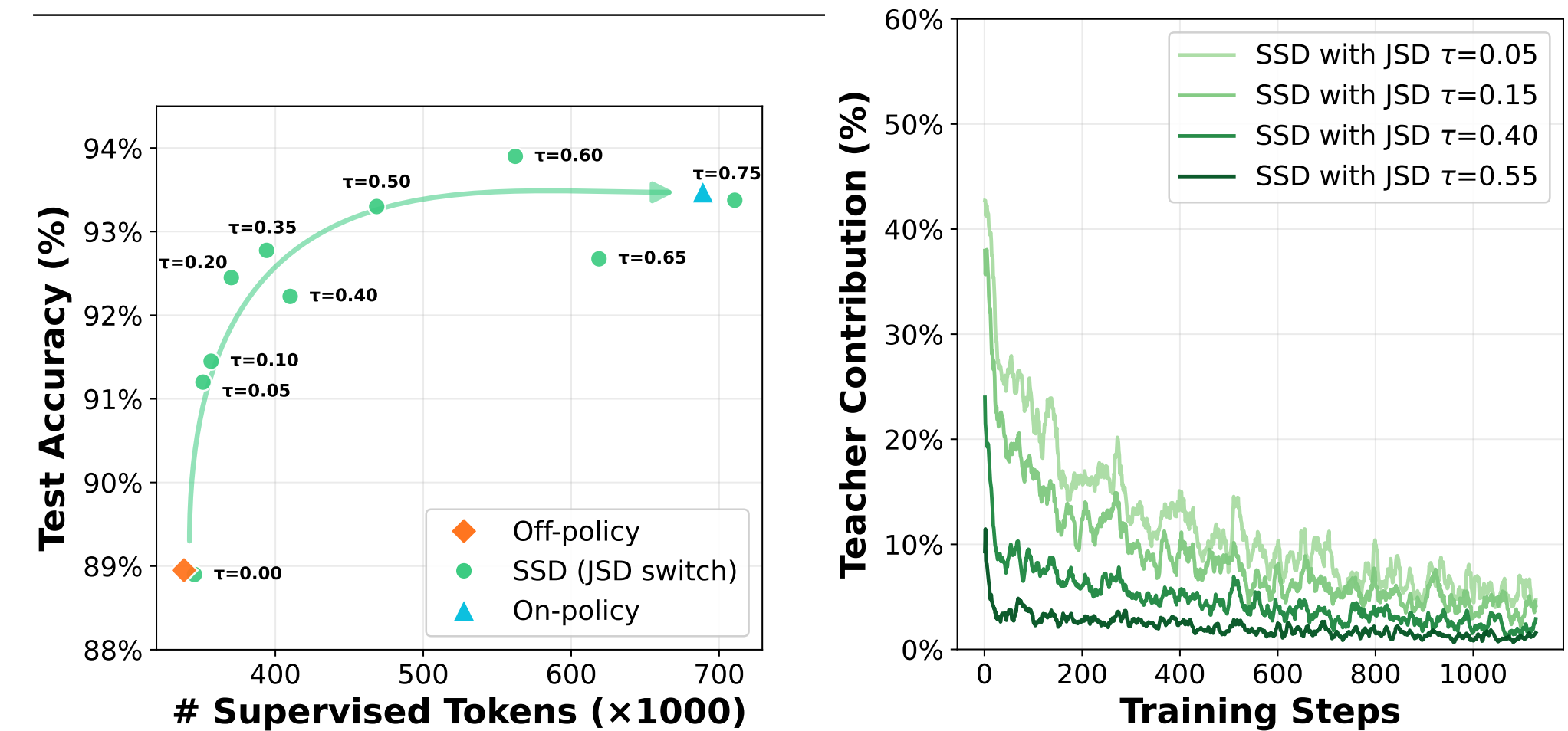
Choosing τ , and a Built-in Curriculum

Figure 5. Accuracy vs. cost across τ (Company Memo).

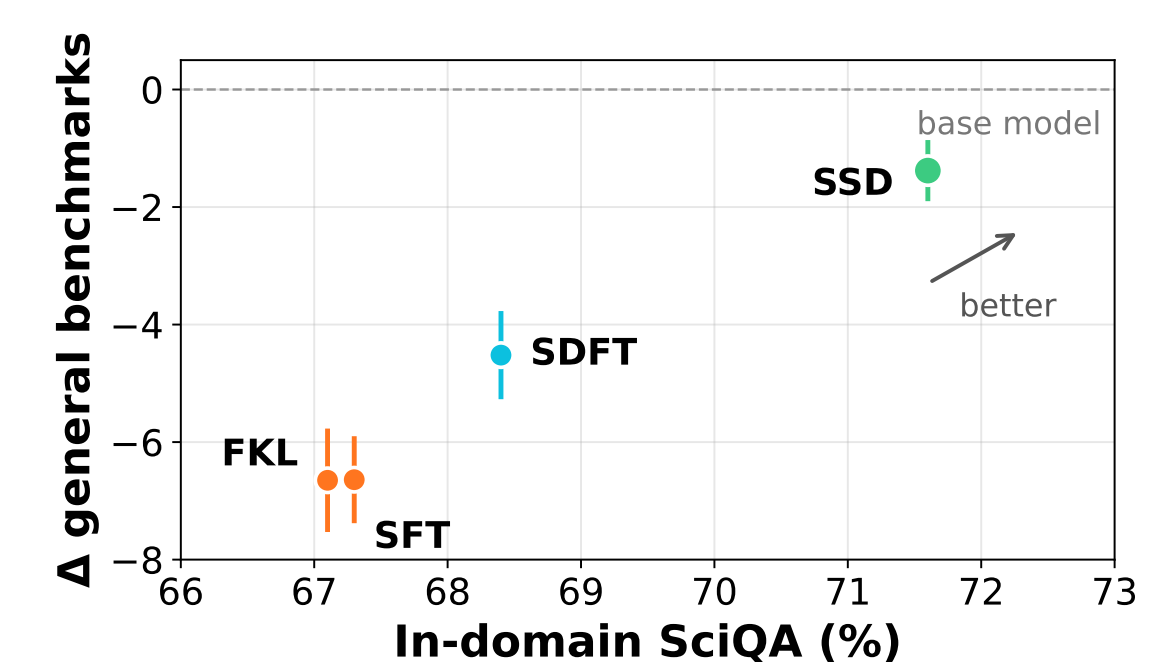
Figure 6. Share of rollout tokens written by the teacher.

- Sweeping τ traces a **Pareto frontier**; $\tau \in [0.2, 0.5]$ (JSD) works well.
- The teacher writes up to 40% of tokens early on; its share falls as the student learns, so training becomes **on-policy without a schedule**.

Less Forgetting

Qwen2.5-7B on SciKnowEval chemistry, scored on six general benchmarks.

SSD has the best SciQA (**71.6**) and the smallest drop (**-1.4** vs **-4.5** for SDFT).



Conclusion

- The useful signal in self-distillation is concentrated at the few tokens where the document changes the prediction.
- Letting the teacher write only there gives **on-policy** accuracy at close to **off-policy** cost, with less forgetting.